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Patent Highlight

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Inhibitors of Adaptor-Associated Kinase 1 (AAK1) May Treat Neuropathic Pain, Schizophrenia, Parkinson's Disease, and Other Disorders

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Patent Application Title:	Biaryl Kinase Inhibitors		
Patent Application Number:	WO 2017/059085 Al	**Publication date:**	6 April 2017
Priority Application:	IN 3170/DEL/15	**Priority date:**	1 October 2015
Inventors:	Bronson, J. J.; Chen, L.; Ditta, J. L.; Dzierba, C. D.; Jalagam, P. R.; Luo, G.; Macor, J. E.; Maishal, T. K.; Nara, S. J.; Rajamani, R.; Sistla, R. K.; Thangavel, S.		
Applicants:	Bristol-Myers Squibb Company; Route 206 and Province Line Road, Princeton, New Jersey 08543, USA.		
Disease Area:	Schizophrenia, Parkinson's disease, neuropathic pain, bipolar disorder, and Alzheimer's disease	**Biological Target:**	Adaptor associated kinase 1 (AAK1)

Summary:

The invention in this patent application relates to biaryl compounds represented generally by formula (I), which can inhibit the adaptor-associated kinase 1 (AAK1). These compounds may provide useful treatments for disorders such as neuropathic pain, Alzheimer's disease (AD), Parkinson's disease, and schizophrenia.

One of the essential cellular processes is endocytosis. It is a mechanism by which molecules such as proteins, which are too large to pass through cell membranes, can be transported (or internalized) into the inside of the cells. The process of endocytosis in mammalian cells involves the use of specific clathrin-coated pits on the cell membranes that are characterized by a unique triskelion-shape structural lattice. This lattice is made by the polymerization of cytosolic clathrin protein onto the cell membrane. The large molecule (the cargo) is "packaged" into these pits. Then the clathrin-coated pits are internalized to form clathrin-coated vesicles. Subsequently, the vesicles will bud inside the cell from the plasma membrane with their cargos, which are delivered into their cellular destinations.

Recent studies on the inhibition of clathrin-mediated endocytosis in an AD mouse model suggest a role for this process in amyloid β (Aβ)-induced collapse of growth cone that leads to axonal degeneration and memory impairment. The inhibition of clathrin-mediated endocytosis was found to prevent amyloid β-induced axonal damage and thus may potentially be beneficial in treating AD.

The Ark1/Prk1 family of serine/threonine kinases initiate phosphorylation cycles that control the endocytic process in mammalian cells. Members of this family include cyclin-G-associated kinase (GAK) and adaptor-associated kinase 1 (AAK1). These proteins are characterized by containing homologous kinase domains, but they also contain other nonhomologous functional domains.

AAK1 modulates the process of clathrin-coated endocytosis. AAK1 is so named because it associates with the adaptor protein complex 2 (AP-2). AP-2 is a heterotetramer, which contains two large subunits (α and $\beta 2$), a medium subunit ($\mu 2$), and a small subunit ($\sigma 2$). It links receptor cargo to the clathrin coat. The binding of clathrin to AAK1 stimulates AAK1 kinase activity. Stimulated AAK1 phosphorylates the $\mu 2$ subunit of AP-2 to promote its binding to tyrosine-containing sorting motifs on cargo receptors. While the phosphorylation of $\mu 2$ is not required for receptor uptake, it enhances the efficiency of the internalization process.

AAK1 has been linked to several disorders and diseases as highlighted in the following:

- AAK1 was identified as a potential therapeutic target for the treatment of neuropathic pain. Recent research has shown that AAK1 knockout mice exhibit a high resistance to pain. Therefore, the inhibition of AAK1 may be beneficial in treating neuropathic pain.
- Researchers have identified AAK1 as an inhibitor of Neuregulin-1 (NRG1)/ErbB4 (a receptor tyrosine-protein kinase) signaling in PC12 cells. They observed that either RNA interference-mediated gene silencing or treatment with K252a (a known inhibitor of AAK1 kinase activity) can cause a loss in AAK1 expression and that in turn results in the potentiation of NRG1-induced neurite outgrowth. These treatments also cause increased ErbB4 expression and its accumulation in or near the plasma membrane. NRG1 and ErbB4 are putative schizophrenia susceptibility genes. Single-nucleotide polymorphisms (SNPs) in both genes have been linked to multiple schizophrenia endophenotypes. Studies have also revealed that NRG1 and ErbB4 KO mouse models have shown schizophrenia relevant morphological changes and behavioral phenotypes.
- A single nucleotide polymorphism in an intron of the AAK1 gene has been associated with the age of onset of Parkinson's disease.

These findings suggest that inhibition of AAK1 activity may be a viable therapeutic target to develop treatments for schizophrenia, cognitive deficits in schizophrenia, Parkinson's disease, neuropathic pain, bipolar disorder, and possibly Alzheimer's disease. The compounds of formula I described in this patent application are inhibitors of AAK1 and may potentially be used as therapeutic agents to treat these disorders.

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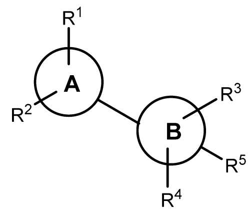
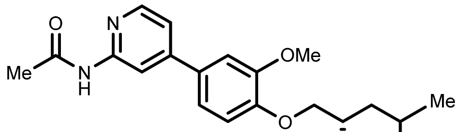
Important Compound Classes:



Formula (I)

Key Structures: The inventors reported the structures and synthesis procedures of 360 examples of formula (I) including the following representative examples:



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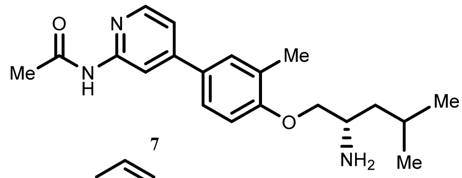

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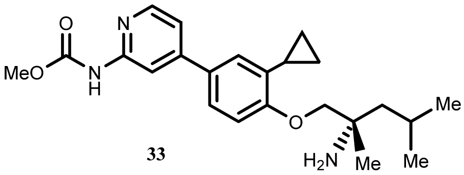

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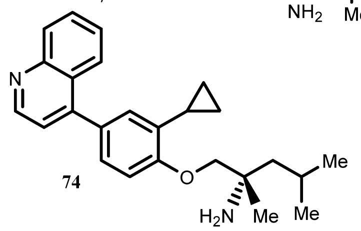

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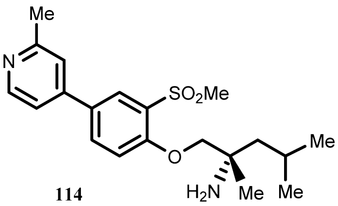

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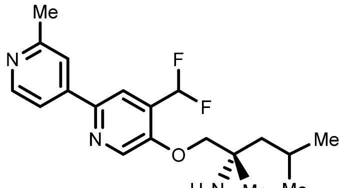

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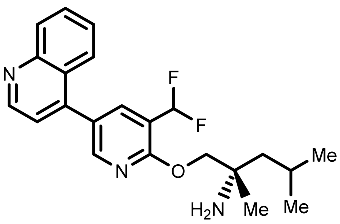

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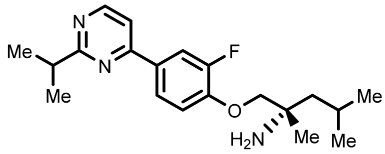

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Biological Assay:

- AAK1 Kinase Assay
- AAK1 Knockout Mice

Biological Data: The inventors reported the IC_{50} data for inhibition of AAK1 obtained from testing the compounds of formula I using the AAK1 kinase assay. The data from testing the above representative examples are included in the following table:

Compound	AAK1 IC_{50} (nM)	Compound	AAK1 IC_{50} (nM)
2	3.0	**114**	1400
7	0.42	**122**	0.51
33	0.32	**204**	0.23
74	0.07	**337**	886

Recent Review Articles:

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Notes
The author declares no competing financial interest.

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